

CO5 – AT2 – SHORT ANSWERS TEST

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Course Name: Data Structure
Course Code: CSA0307

QUESTION 1

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QUESTIONS

Q1 Define a graph. 2 marks

Q2 What are the major types of graphs? 2 marks

Q3 Differentiate between directed and undirected graphs. 2 marks

Q4 What is a weighted graph? 2 marks

Q5

Q1 Define a graph. 2 marks

Define a graph.

Your Answer

Graph is Data Structure Consisting vertices (nodes) and edges to Connect them.

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QUESTION 2

QUESTIONS

Q1 Define a graph. 2 marks

Q2 What are the major types of graphs? 2 marks

Q3 Differentiate between directed and undirected graphs. 2 marks

Q4 What is a weighted graph? 2 marks

Q2 What are the major types of graphs? 2 marks

What are the major types of graphs?

Your Answer

There two major types.
1. Directed Graph
2. Undirected Graph

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QUESTION 3

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QUESTIONS

Q1 Define a graph. 2 marks

Q2 What are the major types of graphs? 2 marks

Q3 Differentiate between directed and undirected graphs. 2 marks

Q4 What is a weighted graph? 2 marks

Q5

Q3 Differentiate between directed and undirected graphs. 2 marks

Differentiate between directed and undirected graphs.

Your Answer

In directed Graph Edges has Direction, While an Undirected Graph Edges has No Direction

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QUESTION 4

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QUESTIONS

Q1 Define a graph. 2 marks

Q2 What are the major types of graphs? 2 marks

Q3 Differentiate between directed and undirected graphs. 2 marks

Q4 What is a weighted graph? 2 marks

Q5

Q4 What is a weighted graph? 2 marks

What is a weighted graph?

Your Answer

A weighted Graph is a Graph each edges has weight or cost Associated it

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QUESTION 5

QUESTIONS

Q1

Define a graph.
2 marks

✓

Q2

What are the major types of graphs?
2 marks

✓

Q3

Differentiate between directed and undirected graphs.
2 marks

✓

Q4

What is a weighted graph?
2 marks

✓

Q5

What is a topological sort?
2 marks

✓

Q5

What is a topological sort?

2 marks

What is a topological sort?

Your Answer

Topological sort is linear ordering of Directed Graph Such That Every Edge A comes Before B

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QUESTION 6

QUESTIONS

Q1

Define a graph.
2 marks

✓

Q2

What are the major types of graphs?
2 marks

✓

Q3

Differentiate between directed and undirected graphs.
2 marks

✓

Q4

What is a weighted graph?
2 marks

✓

Q5

What is a topological sort?
2 marks

✓

Q6

A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.

2 marks

A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.

Your Answer

Valid Topological ordering : A,B,C,D,E,F
Explanation : Because Here B to C Satisfy , C to D Satisfy and E to F Satisfy so This Order is Valid.

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QUESTION 7

QUESTIONS

Q1

Define a graph.
2 marks

✓

Q2

What are the major types of graphs?
2 marks

✓

Q3

Differentiate between directed and undirected graphs.
2 marks

✓

Q4

What is a weighted graph?
2 marks

✓

Q5

What is a topological sort?
2 marks

✓

Q7

An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E, D-F, and E-F. Starting from A, determine the shortest path to F using BFS.

2 marks

An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E, D-F, and E-F. Starting from A, determine the shortest path to F using BFS.

Your Answer

Starting With A
level 0 : A
level 1 : B,C
level 2 : D,E
level 3 : F
Shortest path : A - B - D - F

Save Answer